

InventOR: A Human-Reviewed, Prompt-Configured Large Language Model Decision Support System for Material-Level Inventory Control

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Abstract

Material planners must set a reorder point, a safety stock and an order quantity for thousands of plant-material pairs, usually under ERP parameters that were last revised long ago. We define and evaluate *InventOR*, a decision support system (DSS) for this recurring decision. For each material, a large language model (LLM) configured only through a prompt, without training or fine-tuning, proposes a reorder-point/order-quantity (r, Q) policy with a written justification and an escalation flag. Deterministic software gates the proposal against minimum order quantities and storage limits and replays it on held-out demand before the planner, who remains the decision-maker, reviews it. We derive six design requirements and evaluate the artifact on 365 plant-material pairs from three plants of an industrial manufacturer, using three independent LLM runs, seven reference arms and an ablation that withholds the ERP planning parameters. In a hard-limit storage replay, the LLM policies cost about 11% less than ERP-derived parameters at a mean fill rate 0.1–0.4 points lower, but 6%, 8% and 18% more than

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history-only, Croston/SBA and empirical-quantile (r, Q) baselines. The saving comes from carrying 13–17% less inventory with 11–20% more stockout days; when shortages are priced at one unit cost, ERP-derived parameters are cheaper and a simulation-tuned policy is cheapest. Static storage checks are insufficient: simulated trajectories exceed storage limits for 53–66% of pairs. InventOR is a checkable, human-reviewed candidate generator, not an optimizer that beats data-driven baselines; benefits to planners are untested. We contribute design principles, review rules and a research agenda.

Keywords: Decision Support Systems, Large Language Models, Generative Artificial Intelligence, Inventory Control, Human–AI Decision Making, Design Science, Operations Research

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point, a safety stock and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand, supplier performance and storage constraints change. Revisiting them requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. A conversational large language model (LLM) with a code interpreter appears to offer this scarce skill, and a growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5].

Three problems limit its use. First, LLM outputs can be unfaithful or fabricated [6], so a proposed policy is a candidate, not a decision. Second, superiority claims over ERP parameters are hard to interpret when the LLM and the comparators receive different information. Third, a storage rule that

15 holds when a policy is written down can be violated as stock builds up over
16 time.

17 This paper treats the problem as one of decision support system (DSS)
18 design. We define *InventOR* as a human-reviewed, model-driven DSS for
19 the recurring decision of setting replenishment parameters per plant-material
20 pair (Section 2). A planner remains the decision-maker. The system supplies,
21 for each material, a candidate (r, Q) policy proposed by an enterprise LLM
22 application that is configured only through a prompt and receives no project-
23 specific training or fine-tuning, together with a justification, an escalation
24 flag and a counterfactual simulation; deterministic software parses, gates
25 and simulates every proposal before the planner reviews it and alone decides
26 whether any ERP parameter is changed.

27 We follow a design-science logic [7, 8]: we state problem and scope, de-
28 rive design requirements, build the artifact and evaluate it on real industrial
29 data. The questions are how an LLM should be embedded so that its pro-
30 posals are bounded, verifiable, reproducible and reviewable; what cost and
31 service they achieve when the information available to each approach is con-
32 trolled; whether they respect storage limits along simulated trajectories and
33 are stable across runs; and when a planner should trust, escalate or reject a
34 proposal.

35 Our contributions are: (C1) a definition and scope of the DSS with six
36 design requirements that bound the role of the LLM (Section 2); (C2) an
37 architecture and operating rules that separate generation, deterministic gat-
38 ing, simulation and planner review, with design principles (Sections 4 and 8);
39 (C3) an information-controlled, trajectory-level evaluation with seven refer-
40 ence arms, an ablation denied the ERP planning parameters and a hard-
41 limit storage replay (Section 5); and (C4) complete-cohort evidence from

three LLM runs on 365 plant-material pairs, with managerial and research implications.

Our main findings are as follows. Against ERP-derived parameters the LLM policies are about 11% cheaper at a mean fill rate 0.1–0.4 points lower, but the LLM costs 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile (r, Q) baselines at essentially equal service, and its saving over ERP comes from carrying less inventory at the price of more stockout days. Storage limits that hold in a static check are violated along unconstrained trajectories for 53–66% of pairs. The simulation therefore does not show superior optimization by the LLM; it shows that a prompt-configured LLM can produce checkable candidate policies without code or training, whose benefit to planners remains to be tested.

2. Problem, Scope and Definition

2.1. The decision problem

The decision supported is the periodic (re)setting of a reorder point, a safety stock and an order quantity for each plant-material pair held in an ERP system. In the classification of Gorry and Scott Morton [9] it is a partly structured operational-control decision: the calculation is textbook, but the choice of estimator, the treatment of intermittent or trending demand, the plausibility of recorded lead times and the handling of constraints require judgement that is applied inconsistently when thousands of materials compete for a few analysts' time. The decision-maker is the material planner; hard constraints are the minimum order quantity (MOQ) and a physical storage limit; and feedback is slow, because a changed parameter shows its effect only in later demand. The scope is one pair at a time (single item, single

echelon, continuous review, lost sales, (r, Q) family); order release, supplier selection, forecasting-model development and multi-echelon networks are out of scope, and the LLM is a candidate generator that does not forecast, place orders, simulate or decide.

2.2. Definition and level of automation

Definition. *InventOR is a human-reviewed, model-driven decision support system that, for one plant-material pair at a time, (i) obtains from a large language model, configured only by a written prompt, a candidate (r, Q) policy with a justification and an escalation flag; (ii) verifies the candidate deterministically against explicit constraints; (iii) evaluates it by counterfactual simulation against reference policies on held-out demand; and (iv) presents the result to a planner, who alone decides whether ERP parameters are changed.*

In the classical DSS architecture of data, model and dialog components [10, 11], the LLM and the simulator form the model component, the material-scoped data extract and the prompt form the data and knowledge components, and the review interface is the dialog component (Section 4.1). In terms of the levels of automation of Parasuraman et al. [12], InventOR operates at a low-to-intermediate level: it selects and suggests one alternative and leaves execution to the human. Accepting automated suggestions invites complacency and automation bias [13], which the review rules of Section 4.3 are meant to counter.

90 2.3. Design requirements

91 The requirements follow from the known unreliability of LLM outputs
 92 [6], guidance on human–AI interaction and accountability [14, 15], run-to-
 93 run nondeterminism of LLM inference [16] and the constraints of inventory
 94 control. Table 1 ties each to a design feature (Section 4) and an evaluation
 95 (Section 6).

Table 1: Design requirements, the design features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.8
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.7
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

96 3. Related Work and Positioning

97 *Decision support systems and human–AI decision making.* DSS research
 98 has long distinguished data-, model- and knowledge-driven systems and has
 99 stressed that the value of a DSS lies in improving the judgement of a hu-
 100 man decision-maker rather than in replacing it [9, 10, 11]. Research on
 101 human–automation interaction shows that the way suggestions are scoped,
 102 presented and verified determines whether users rely on them appropriately
 103 or over-rely on them [12, 13],. Recent DSS work derives design principles
 104 for LLM-based assistants [17], reports design and pilot implementations of
 105 AI-enabled decision support in operations [18], and shows that the uptake of

106 algorithmic advice depends on transparency and on the user’s metacognition
107 [19, 20]; evaluations of data-driven forecasting systems in practice likewise
108 stress lessons beyond accuracy [21]. Model-driven support also shapes how
109 groups handle disagreement [22], and human override of algorithmic safety-
110 stock advice is a documented risk [23]; InventOR therefore keeps the planner
111 as decision-maker and makes the evidence for each proposal explicit.

112 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
113 map the opportunities for generative AI in supply chains [1, 2, 24, 25, 3, 4, 26]
114 and stress that adoption depends on assigned human roles, data quality and
115 governance [5, 27, 28, 29, 30]. Evidence from manufacturing planning is thin:
116 a fine-tuned LLM for digital-manufacturing optimization [31], a transformer
117 trained to set order timing and quantity and benchmarked against (r, Q) [32],
118 and experiments showing that ChatGPT reproduces human decision biases
119 of managers [33]. Most technical LLM work translates problem statements
120 into optimization models and solver code [34, 35, 36, 37], studies multi-agent
121 control in inventory flows [38, 39], disruption modelling [40] or classical OR
122 tasks such as routing [41]. The closest LLM study to ours uses an LLM
123 as a forecasting copilot for spare-parts demand, combined with simulation,
124 and reports high variability for irregular demand that can be reduced by
125 averaging repeated queries and enriching prompts [42]. We test run-to-run
126 variability directly (Section 6.7); the difference is that our LLM proposes a
127 complete policy, not a forecast.

128 *Data-driven inventory control without LLMs..* Learning-based replen-
129 ishment is well established, including data-driven policies learned from
130 sequentially observed non-stationary demand [43] and forecasts augmented
131 with leading indicators [44], and heuristics remain competitive in constrained

132 settings such as perishable inventory [45]. Deep reinforcement learning has
 133 been applied to omnichannel and multi-echelon systems [46, 47, 48] and
 134 aerospace replenishment [49]; deep-learning forecasting has been integrated
 135 with replenishment [50]; contextual distributionally robust optimization
 136 uses side information to set order quantities [51]; and distribution-free
 137 online algorithms handle capacitated warehouses under uncertain price and
 138 demand [52]. The closest operational neighbour is a trained data-driven
 139 framework that jointly sets replenishment parameters, safety stock and
 140 safety lead time on real automotive data [53]. Reinforcement learning has
 141 also been combined with demand forecasting to optimize order points [54].
 142 These methods need training or tuning; InventOR needs none, but its
 143 generator cannot be inspected beyond its written justification.

144 *Intermittent and spare-parts demand..* Material histories in manufacturing
 145 are frequently intermittent or lumpy. The statistical literature offers special-
 146 ized estimators and classification schemes [55, 56, 57, 58, 59], and recent stud-
 147 ies develop feature-based forecast combinations and temporal aggregation for
 148 intermittent demand [60, 61], forecasting for spare parts across product life-
 149 cycle phases [62], and integrated forecasting and planning or multi-echelon
 150 stocking models for repairable and spare parts [63, 64].

151 InventOR is not a new inventory model and does not claim to beat
 152 trained or optimized methods; its (r, Q) policy is the standard workhorse
 153 of continuous-review control [65, 66, 67].

154 4. The InventOR Decision Support System

155 4.1. Architecture and components

156 Following the classical DSS components [10, 11], InventOR has a *data*
157 component (a material-scoped extract of daily demand, lead times, working-
158 day flags, cost and constraint parameters and ERP reference values; the
159 fields supplied are controlled and logged; DR4), a *knowledge* component (the
160 system prompt file defining the task, the prohibited methods, the storage
161 rules and the output schema, Supplementary File S1; DR1, DR5), a *model*
162 component split into a generator (an enterprise LLM application with a code
163 interpreter, configured only by the prompt, proposing one (r, Q) policy with
164 a justification and an escalation flag; DR1) and a deterministic verifier and
165 simulator (parser, gate, held-out simulation with hard storage-limit replay
166 and reference arms; DR2, DR3), a *dialog* component (policy, justification,
167 flag and simulated comparison shown to the planner under the trust, escalate
168 and reject rules; DR6) and a *governance* layer (frozen prompt file, per-call
169 log of input hash, time, session and raw response, repeated runs; DR5). The
170 generator is the only non-inspectable component. In the evaluated prototype
171 the dialog component is the per-pair output record; no interactive interface
172 was built or evaluated.

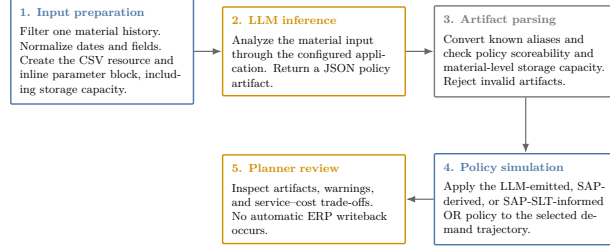
173 4.2. Operation in four steps

174 For one material, InventOR answers how much safety stock to hold, at
175 what level to reorder and how much to order, in four steps (Figure 1).

176 **Step 1. Generate.** The LLM, accessed through an enterprise application
177 with a code interpreter, receives one material’s history (daily de-
178 mand, realized lead times, working-day flags) and a short parameter

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [68] instructs it to compute demand and lead-time statistics, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material’s demand observed *after* the data cutoff, a held-out evaluation in the

196 sense of out-of-sample forecast testing [69, 70], together with the
 197 comparator policies, producing holding and ordering cost, fill rate
 198 and stockout days.

199 **Step 4. Review.** A planner sees the policy, the LLM’s justification and
 200 flag, and the simulated comparison. Only the planner changes ERP
 201 parameters.

202 The LLM is a *candidate generator* whose output is never trusted without
 203 the gate; it does not forecast, place orders or run the simulation. The prompt
 204 prohibits Monte Carlo simulation, distribution fitting, machine-learning li-
 205 braries and specialized intermittent-demand estimators, so that the policy
 206 rests on transparent statistics.

207 4.3. *Trust, review and reject rules*

208 We propose three rules for planners, in line with guidelines for human–AI
 209 interaction and AI auditing [14, 15]. The rules and the 0.5-point threshold
 210 are design proposals not evaluated with planners; the trust rule would be
 211 met by a minority of pairs (the LLM is cheaper than the history-only OR in
 212 74–77 of 346).

213 **Trust (apply after review)** when the proposal passes the gate, the sim-
 214 ulated trajectory respects the storage limit when orders are truncated at it,
 215 and the simulated cost is not above that of the best history-only OR ref-
 216 erence (Croston/SBA or empirical quantile) with a fill rate not more than
 217 about 0.5 points lower.

218 **Escalate** when the LLM raises an escalation flag, the material is lumpy or
 219 has strongly seasonal or trending demand, the plant or calendar data were
 220 repaired, or the policy differs strongly between repeated generations.

221 **Reject** when the proposal fails the gate after the retry budget, exceeds the
222 storage limit in the hard-limit replay, or is dominated (higher cost and lower
223 fill) by a history-only reference.

224 *4.4. Provenance and reproducibility of the configuration*

225 The system prompt, a plain-text file deployed unchanged in the enter-
226 prise application, defines the task, forbidden methods, capacity rules and
227 output schema (Supplementary File S1). Every call logs the SHA-256 of its
228 input, UTC time, session identifier and raw response. The platform does
229 not expose the model identifier or sampling settings; the evaluated model
230 (Anthropic Claude Sonnet 4.6 with a code interpreter) is operator-declared.
231 Reproducibility rests on the prompt file, the call logs and three independent
232 runs (Section 6.7).

233 **5. Application and Evaluation Design**

234 *5.1. Evaluation approach*

235 The evaluation addresses the design requirements (Table 1). The pri-
236 mary basis is the hard-limit replay with working-day receipts and no short-
237 age penalty; sensitivity analyses vary these choices. Following design science
238 [7, 8], the evaluation is artificial and ex post in the framework of Venable
239 et al. [71]: real data and a faithful simulation but no live users. It estab-
240 lishes technical efficacy, feasibility and reliability, not usability, adoption or
241 improved planner decisions.

242 *5.2. Industrial application and data*

243 The application is a pilot at a large manufacturer. The dataset contains
244 233,078 daily records for 411 plant-material pairs from three plants (A, B and

245 C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed
246 to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring (234 in
247 Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19
248 pairs (all in Plant A) the MOQ plus the reference safety stock exceeds the
249 storage limit, so no comparator can satisfy both rules; these are excluded
250 from outcome comparison, leaving a *common-feasible cohort* of 346 pairs
251 (215, 65 and 66). Demand is concentrated (the five largest materials account
252 for 40.3% of demand). The source working-day flag of Plant C is all zero; we
253 reconstruct it from the same-date majority of Plants A and B and test this
254 repair in Section 6.8. Data quality is a known threat to decision support [72]
255 and inventory records [73, 74, 75].

256 Each pair is sent as a material-scoped CSV with daily demand, recorded
257 lead times, a working-day flag, the minimum order quantity, unit cost,
258 holding-cost rate, ordering cost, service-level target, the storage limit, and
259 two *ERP planning parameters* (current safety stock and safety lead time),
260 which are withheld in the ablation arm. The CSV also carries supplier and
261 lane context fields that the prompt does not use.

262 5.3. Reference arms and information content

263 ERP safety stock and safety lead time embed planner judgement and are
264 alternative ways of buffering unreliability [76], and planned lead times can
265 drift upward [77]. The LLM and the classical comparators therefore do not
266 necessarily use the same information, and we construct arms that make it
267 explicit (Table 2).

268 Three further arms use no ERP input and represent stronger data-driven
269 practice. The *Croston/SBA OR* replaces the mean daily demand by the
270 Croston estimate with the Syntetos–Boylan bias correction [55, 56] (smooth-

Table 2: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM’s full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None		Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None		Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None		Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

ing constant 0.1), uses the root mean squared one-step forecast error as
 demand spread and the history-only lead-time moments, which suits inter-
 mittent demand [57]. The *empirical-quantile OR* sets the reorder point to the
 service-level quantile of historical rolling lead-time demand over the rounded
 mean lead time. The *simulation-tuned (r, Q)* policy selects, from a grid of
 reserve factors and order-quantity multiples around the history-only policy,
 the cheapest candidate whose fill rate on the last 250 working days *before*
the cutoff meets the service target (or the best-fill candidate if none does).

279 No post-cutoff data are used in any of these arms. We do not implement a
 280 deep reinforcement-learning baseline [47, 46]; this is a limitation (Section 8).

281 5.4. *Simulation, capacity replay and metrics*

282 Each pair is simulated independently on the post-cutoff horizon as a
 283 single-item, single-echelon continuous-review (r, Q) system [65, 78, 79, 67, 66]
 284 with lost sales [80], in the spirit of simulation-based experiments on supply
 285 and timing uncertainty [81], a shared opening stock, an empty pipeline and
 286 arrivals after the rounded mean lead time in working days (Appendix Ap-
 287 pendix A). MOQs are treated as hard order-size constraints, as imposed
 288 by suppliers [82], and storage limits as capacity bounds on the replenish-
 289 ment decision [52]. The primary metrics are the mean material-horizon
 290 holding-plus-ordering cost, the mean material-level fill rate, the aggregate
 291 (volume-weighted) fill rate and the number of stockout days. We report
 292 paired differences with 10,000-sample bootstrap 95% intervals.

293 A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage
 294 limits hold as inventory evolves, because the reorder point includes lead-time
 295 demand and inventory on order can exceed the limit on arrival. We therefore
 296 record the post-receipt on-hand stock for every policy and day and report
 297 the share of pairs and days above the limit (the *unconstrained* simulation),
 298 and we *replay* every policy with each order truncated to the storage limit
 299 minus the current inventory position (the *hard-limit replay*), which is our
 300 primary evaluation.

301 5.5. *Repeated runs and sensitivities*

302 To examine run-to-run variation, the entire cohort is generated three
 303 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1

304 serves as the reference for tabulated results unless stated otherwise; runs 2
305 and 3 provide replicates. Sensitivity analyses vary the working-day recon-
306 struction (majority, any-plant, strict-all, exclusion of Plant C and calendar-
307 day arrivals), the shortage penalty (zero, one and five times unit cost).

308 6. Results

309 6.1. Completion and policy characteristics

310 All four runs (three with the full input and one ablation, Section 6.4)
311 return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most
312 three attempts per pair. The validity gate rejects non-positive reorder points
313 and order quantities, negative safety stocks and reorder points below the
314 safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of
315 the 346 in the cohort below), so for a majority of pairs the SAP-derived arm
316 holds no safety stock at all.

317 6.2. Headline comparison with ERP-derived parameters

318 Unless stated otherwise, all results use the hard-limit replay on the
319 346-pair common-feasible cohort with working-day receipts and no shortage
320 penalty (Section 6.5 gives the unconstrained simulation). Table 3 compares
321 all arms; its last columns give the LLM cost relative to each arm in the three
322 runs (positive: the LLM costs more). In run 1 the LLM policies cost 11.4%
323 less than SAP-derived policies and 30.5% less than the SLT-informed OR
324 (mean 2,928 versus 3,305 and 4,211), at a mean fill 0.18 points lower than
325 SAP-derived (98.11% versus 98.30%; interval -0.54 to $+0.10$). The mean
326 cost difference against SAP-derived is -377 (95% interval -651 to -154);
327 the LLM is cheaper in 241 of 346 pairs and, in 226, simultaneously no more
328 expensive and no worse in fill. It incurs more stockout days (910 versus 804).

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same number of orders per year (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.8 shows that once shortages carry a cost this trade no longer favours the LLM.

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	801	−11.4 / −11.3 / −11.4
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	−30.5 / −30.4 / −30.5
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	−31.1 / −31.0 / −31.1
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	+5.9 / +6.0 / +5.9
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	+8.2 / +8.4 / +8.3
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	+17.7 / +17.8 / +17.7
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	−10.4 / −10.3 / −10.3 [‡]
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	−8.8 [†] (vs SAP)

[†] Other comparisons of the ablation are in Section 6.4. [‡] Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.8).

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mainly measures how heavily ERP inputs over-buffer.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with fill differences of −0.03, −0.25 and +0.01 points whose intervals include zero, and is cheaper in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97

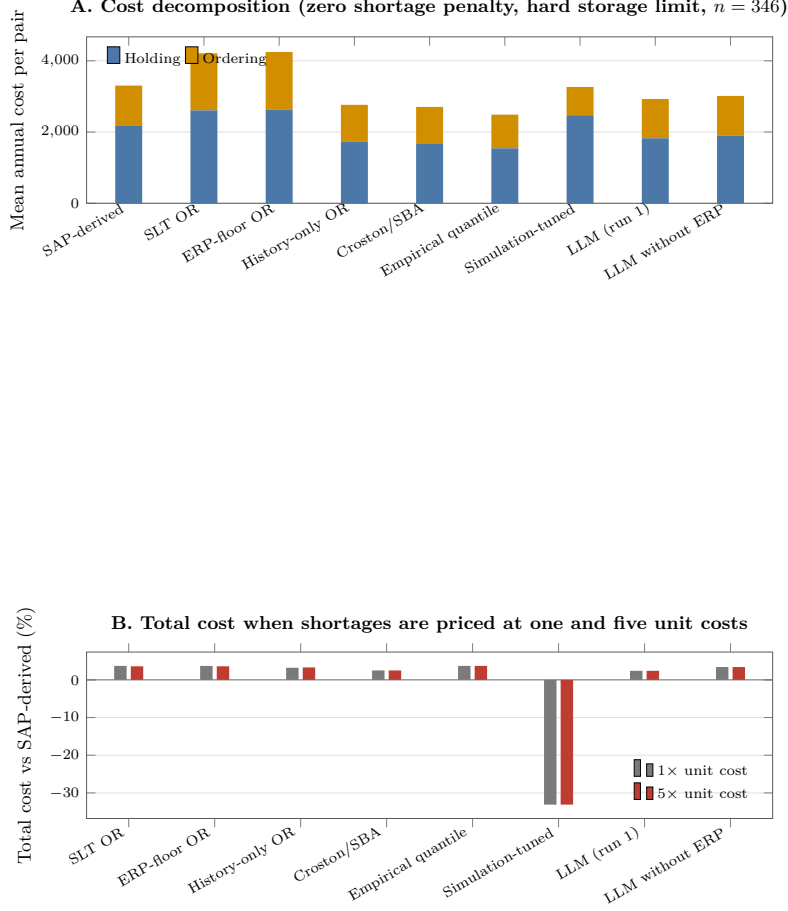


Figure 2: (A) Mean holding and ordering cost per pair for each arm, hard-limit replay, zero shortage penalty (346 pairs). (B) Mean total cost including shortages priced at one and five times the unit cost, relative to SAP-derived parameters. Absolute values are in Table 3 and in the text of Section 6.8.

to +362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489, fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM’s advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean

fill of 96.05% and 1,973 stockout days, because it is tuned on the pre-cutoff year. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher mean fill, but the interval includes zero (−2,527 to +857); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.8).

Third, the LLM improves on ERP-derived parameters (11.3–11.4% cheaper at 0.14–0.40 points lower fill; the interval excludes zero only in run 2) but does not reach the best classical baselines, which also yield a policy for every material at lower cost.

6.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives the same inputs as in runs 1–3, but the ERP safety stock and safety lead time are blanked in the CSV and marked “not supplied”; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters. All 365 pairs returned a valid, storage-compliant policy, and the policies change only slightly beyond ordinary run-to-run variation: the median ratios of safety stock and reorder point to run 1 are 1.00; the reorder point differs by more than 10% in 54.6% of pairs (48.3% and 48.0% between ERP-supplied runs) and the safety stock in 11.0% (9.8% and 9.5%); the mean ratio is 1.04 for safety stock and 1.05 for the reorder point. In the hard-limit replay the cost is 3,014 at a mean fill of 98.06%, against 2,928–2,931 in runs 1–3. It is 8.8% below SAP-derived parameters (interval −566 to −61; fill −0.24 points, interval including zero) and 28.4–29.1% below the SLT-informed and ERP-floor ORs, but 9.0% above the history-only OR (interval +134 to +387; cheaper in 91 of 346 pairs), 11.4% above the Croston/SBA OR, 21.1% above the empirical-quantile OR and 7.7% below the simulation-tuned policy. By plant the pattern is unchanged (−10.8% in Plant A, −21.9% in Plant B, +41.6% in Plant C against +36.2%

in run 1), so the Plant C loss does not stem from reliance on ERP parameters. The information-asymmetry concern therefore does not explain the LLM’s advantage over ERP-derived parameters, and the ERP planning fields appear to be of limited use to the LLM. The ablation withholds only these two fields; unit cost, holding rate, ordering cost, MOQ, service target and storage limit remain available, as they are for all reference arms.

6.5. *Storage limits along simulated trajectories*

All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$, yet in the unconstrained simulation post-receipt on-hand stock exceeds the limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for the SLT-informed OR, because inventory on order and lead-time demand add to stock above $SS + Q$. The static check is thus necessary but not sufficient for any method. Under the hard-limit replay no arm exceeds the limit by more than one unit; enforcing it raises the LLM mean cost from 2,850 to 2,928 and SAP-derived from 3,184 to 3,305 without changing the ranking.

6.6. *Plants and demand regimes*

The average hides a strong plant effect (Figure 3, hard-limit replay, run 1). The LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C (cheaper in 8 of 66); the Plant C loss is similar in the other full runs (+32.2%, +31.5%) and persists under every working-day reconstruction in which Plant C can be scored (Section 6.8). Against the data-driven baselines the LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only OR, 13.1%, 2.3% and 6.2% more than the Croston/SBA OR,

403 and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR in Plants
 404 A, B and C. The SLT-informed and history-only ORs coincide in Plant C
 405 because the ERP safety lead time is zero for all its pairs.

406 By demand regime, the 33 lumpy materials show the largest reduction
 407 against SAP-derived (-25.4% in run 1) and the 313 smooth materials -9.8%
 408 at a 0.24-point lower mean fill. On lumpy materials the service effect is
 409 inconsistent (mean fill $+0.38$ points in run 1, -0.29 in run 2), the history-only
 410 and Croston/SBA ORs cost about 9–10% less than the LLM, and only the
 411 empirical-quantile OR is slightly more expensive (1.2%); with 33 materials
 412 this is indicative only. By terciles of the coefficient of variation, the LLM
 413 costs 15.7%, 7.5% and 12.9% less than SAP-derived (lowest to highest), with
 414 a 0.68-point lower mean fill in the middle tercile.

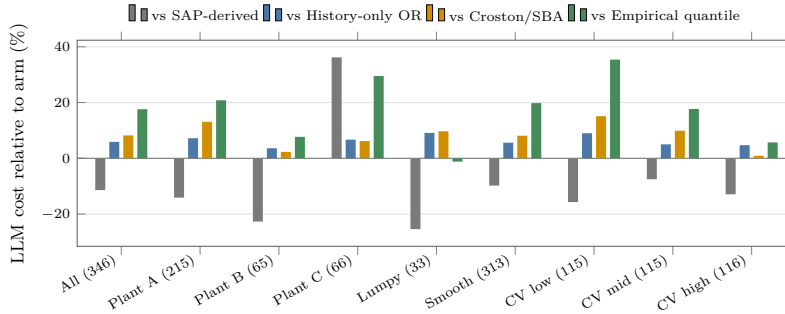


Figure 3: Cost of the LLM (run 1, hard-limit replay, common-feasible cohort) relative to SAP-derived parameters and to three data-driven (r, Q) baselines, by plant, demand regime and coefficient-of-variation tercile (number of pairs in brackets). Positive values mean the LLM costs more. The SLT-informed and history-only ORs coincide in Plant C.

415 6.7. Run-to-run stability

416 Across the three runs the aggregate results are very similar: the LLM
 417 cost relative to SAP-derived is -11.4% , -11.3% and -11.4% , and the policy-
 418 value medians are close (reorder point 1,584, 1,582 and 1,466; order quantity

419 2,737, 2,777 and 2,693; safety stock 683, 688 and 669). Individual policies
 420 vary, however. Between runs 2 and 3 on the 365 pairs, all three values are
 421 identical for only 36 pairs; every policy resolves to the (r, Q) family, and the
 422 median absolute differences are 81 units (reorder point), 6 (order quantity)
 423 and 3 (safety stock). The reorder point differs by more than 10% in about
 424 48% of pairs, and a few pairs vary strongly (one reorder point was 14,882,
 425 14,947 and 29,218 units). Aggregate behaviour is reproducible, but a single
 426 generation per material is a noisy estimate that should be reviewed [16].

427 6.8. Sensitivity analyses

428 *Plant C calendar..* Plant C’s working-day flag required reconstruction, so we
 429 re-scored all policies under four alternatives (any-plant, strict-all, exclusion
 430 of Plant C, calendar-day arrivals). Where Plant C can be scored, the LLM
 431 remains 31–35% more expensive than SAP-derived there in the three full
 432 runs, and the cohort-wide advantage is -10.8% to -11.1% ; the strict and
 433 exclusion variants (280 pairs) show -17.0% to -17.6% . The Plant C result
 434 is therefore not an artefact of the repair, although the LLM saw demand
 435 series built on the repaired calendar.

436 *Shortage penalty..* Figure 2B gives mean total cost per pair (holding, order-
 437 ing and shortage cost) relative to SAP-derived parameters in the hard-limit
 438 replay for penalties of one and five times the unit cost. Costs are dominated
 439 by the shortage term, and the differences among the non-tuned arms (0.1–
 440 3.7%) should be read with caution. The simulation-tuned policy is by far
 441 the cheapest (1,094,043 at one times the unit cost, about 33% below SAP-
 442 derived), because it was selected to meet the service target on the pre-cutoff
 443 history. Among the other arms SAP-derived parameters are the cheapest,
 444 as they carry more inventory and fewer stockouts. LLM run 1 then costs

2.4% more than SAP-derived at both multiples, is about as expensive as the Croston/SBA OR (1,673,003 versus 1,674,925 at one times), and is cheaper than the SLT-informed (1.2%), ERP-floor, empirical-quantile and history-only (0.8%) ORs. The LLM’s cost advantage over ERP-derived parameters is therefore specific to the case in which service shortfalls are not charged.

7. Managerial Insights

1. **Position the tool as a reviewed decision support system, not an autonomous engine.** Single generations differ at the material level (Section 6.7). The gate-simulate-review structure yields a justified candidate for every material and leaves the decision to the planner; whether this helps planners is untested.

2. **Benchmark against data-driven baselines.** ERP-derived parameters and ERP-informed ORs are weak benchmarks (safety stocks are often stale or zero; safety lead times inflate buffers), so comparing only with ERP overestimates the benefit: the LLM cost 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile baselines.

3. **Expect plant-level differences.** Against ERP-derived parameters the LLM was cheaper in Plants A and B (14–23%) and on 33 lumpy materials (about 25%, mixed service) but not in Plant C; we did not test whether escalation rules would catch such cases.

4. **Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$ cannot protect a warehouse: 53–66% of pairs breach the limit in simulation under each of the three arms checked. Enforce a cap in the order-release logic or treat the limit as a design requirement.

469 **5. Price shortages explicitly.** Without a shortage penalty the ranking
470 rewards under-stocking; with a penalty of one unit cost the LLM is about
471 2.4% more expensive than SAP-derived. Set service targets and penalties
472 before reading cost comparisons.

473 **6. Freeze and log the configuration.** Treat the system prompt as a
474 controlled file, log every call (input hash, time, session, raw response)
475 and re-run before changing the prompt.

476 **7. Cost and latency.** Each generation took several minutes, whereas OR
477 baselines take milliseconds; periodic regeneration would support adapt-
478 able planning [83].

479 **8. Discussion, Limitations and Research Perspectives**

480 *What the evidence shows..* A prompt-configured LLM generates feasible, re-
481 viewable (r, Q) policies for every material; in simulation they cost less than
482 ERP-derived parameters but more than data-driven OR policies. InventOR
483 is therefore not presented as outperforming optimization: its role is a re-
484 viewed candidate generator, and whether that helps planners is a hypothesis
485 the simulation cannot test. Against the ERP-informed arms alone the LLM
486 would look far better (-11.4% against SAP-derived, -30.5% against the
487 SLT-informed OR).

488 *Design principles for LLM-based decision support..* We derive six principles
489 from this application and offer them as hypotheses for design-science work
490 [7]. **DP1, bound the generator:** the LLM is a candidate generator whose
491 prompt fixes the method, constraints and output schema, with no authority
492 to act. **DP2, verify deterministically and along time:** check outputs

493 with code and feasibility along the trajectory, because point checks can pass
 494 while constraints are violated over time (Section 6.5). **DP3, show counter-**
 495 **factual evidence:** give the reviewer simulated comparisons with credible
 496 references. **DP4, control information in evaluation:** compare against
 497 baselines with and without the same information and withhold incumbent
 498 inputs in an ablation (Sections 6.3, 6.4). **DP5, freeze and log:** treat the
 499 prompt as a controlled file, log every call and repeat runs (Section 6.7). **DP6,**
 500 **escalate by rule:** define in advance when proposals are trusted, escalated
 501 or rejected, to limit complacency [13] (Section 4.3).

502 *Limitations..* (i) The data come from one manufacturer and three plants. (ii)
 503 The model identifier and inference settings are operator-declared. (iii) No
 504 planner outcomes were measured: the benefits of reviewability, auditability
 505 and planner support and the rules of Section 4.3 are design intentions that
 506 a field study must test. (iv) The simulation uses deterministic arrival at the
 507 rounded mean lead time, ignores observed receipts, corrections and scrap,
 508 starts with an empty pipeline and does not simulate lead-time variability,
 509 which may bias comparisons that depend on safety lead time; we did not
 510 report service-target attainment. (v) The primary results use a zero shortage
 511 penalty; with a positive penalty the ranking changes. Costs are dominated
 512 by a few high-demand materials and the 346 pairs are not independent,
 513 so the bootstrap intervals understate uncertainty. (vi) The (r, Q) family
 514 is single-item and single-echelon. (vii) Individual policies vary across runs,
 515 and Plant C required calendar reconstruction. (viii) We included no deep
 516 reinforcement-learning or neural forecasting baseline, and the ERP-derived
 517 arm (ERP safety stock with our own order quantity) is not an observed
 518 record of practice.

519 *Research perspectives.*

520 1. *Ensembling and hybrids.* Aggregating several generations may reduce
521 individual-policy noise [42], and combining the best data-driven OR policy
522 with LLM review or regime-specific choice [84] could capture the OR cost
523 level and the LLM’s flags.

524 2. *Comparison with trained methods.* Benchmarks against deep
525 reinforcement-learning and other trained policies [53, 46, 47] on
526 identical data and information, and richer policy families such as (s, S)
527 [58, 59], would clarify the trade-off between setup effort and policy
528 quality.

529 3. *Planner-in-the-loop field trials.* Prospective studies should compare plan-
530 ners with and without the aid on matched materials, measuring planner
531 time, override rates and realized service and inventory, and should exam-
532 ine adoption drivers [28, 27, 5].

533 9. Conclusion

534 We defined InventOR as a human-reviewed decision support system for a
535 recurring production-planning decision, the setting of replenishment parame-
536 ters per plant-material pair, and evaluated it against six design requirements
537 on a three-plant industrial dataset. A prompt-configured LLM generated
538 feasible, justified (r, Q) policies for every material without training, and a
539 deterministic gate, a hard-limit simulation and explicit review rules made
540 them checkable. The proposals cost about 11% less than ERP-derived pa-
541 rameters at slightly lower service, but the LLM costs about 6%, 8% and 18%
542 more than history-only, Croston/SBA and empirical-quantile policies; when

lost sales are priced at one unit cost or more, ERP-derived parameters are the cheapest of the non-tuned arms and a simulation-tuned policy is cheapest overall. Storage limits that hold in a static check are exceeded along simulated trajectories for every method, and the LLM loses to ERP at one plant. The evidence supports InventOR as a checkable candidate generator inside a reviewed process, not as an optimizer that beats data-driven baselines; the design principles of Section 8 are hypotheses, and field evidence on planner behaviour and realized outcomes is the central open step.

Appendix A. Formal Definitions

Reference calculation. These are the standard textbook calculations [67, 78]. For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the lead time in working days, K_i the ordering cost, h_i the annual holding cost per unit (unit cost times holding rate), z_i the standard normal quantile of the service target, and 250 the working days per year. The order quantity is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$ and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time for the history-only OR and the mean lead time plus the ERP safety lead time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i . All policies are projected to $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

Simulation. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and ar-

568 rives after the rounded mean lead time in subsequent working observations.
569 Observed receipts, corrections and scrap are ignored. The total cost is
570 $TC_i = h_i \bar{I}_i T / 250 + K_i N_i$, where $\bar{I}_i$ is the mean on-hand stock over the
571 T simulated working days and N_i the number of orders. Fill rate is served
572 divided by demanded units. In the hard-limit replay an order is truncated
573 to `max_storage_unitsi` minus the inventory position, and no order is placed
574 if the truncated quantity is zero.

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582 **Declaration of Competing Interest**

583 The author declares no known competing financial interests or personal
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585 paper.

586 **Data and Code Availability**

587 The raw operational data and material-level artifacts cannot be publicly
588 released because they contain commercially sensitive material, supplier
589 and planning information. A confidentiality-screened release package

590 containing analysis code, the anonymized variable schema, aggregate
591 evaluation results, manuscript sources and prompt files is available at
592 github.com/DressPD/inventor_public. A versioned Zenodo archive of
593 the release is available at <https://doi.org/10.5281/zenodo.22262052>.
594 Supplementary File S1 contains the system prompt used for all runs; the
595 prompt file, the runner code and, for each call, a hash of the input, the
596 UTC time, the session identifier and the raw response are in the release.
597 The model identifier and inference settings are not exposed by the platform
598 and are operator-declared (Anthropic Claude Sonnet 4.6).

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605 **Declaration of Generative AI and AI-Assisted Technologies in the** 606 **Writing Process**

607 During manuscript preparation and revision, the author used OpenAI
608 GPT-5.6 and Anthropic Claude (Sonnet 5.5) models through the OpenCode
609 coding assistant for language editing, restructuring, consistency checks, ci-
610 tation verification against Crossref metadata, LaTeX proofing, and writing
611 and executing the analysis scripts that produce the evidence tables. These
612 assistants are distinct from the Claude Sonnet 4.6 deployment evaluated in
613 the study. The author specified the analyses, reviewed and edited all content

614 and results, and takes full responsibility for the content. No generative AI
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